Mutual Information

An informal introduction

Consider the following two statements

1. Sindhu, an Indian won medal in Olympics, in badminton.
2. India has got one Silver medal in Olympics

- Both the statements have something in common and something not.
  - Common: India won medal in Olympics
  - Unique to 1st statement: Who got? & Which game?
  - Unique to 2nd statement: What medal?

Chain rule for information

$$\log \frac{1}{P(x,y)} = \log \frac{1}{P(x)} + \log \frac{1}{P(y \mid x)}$$

Information content of x and y

Information content of x

Information content of y given x

Chain rule for entropy

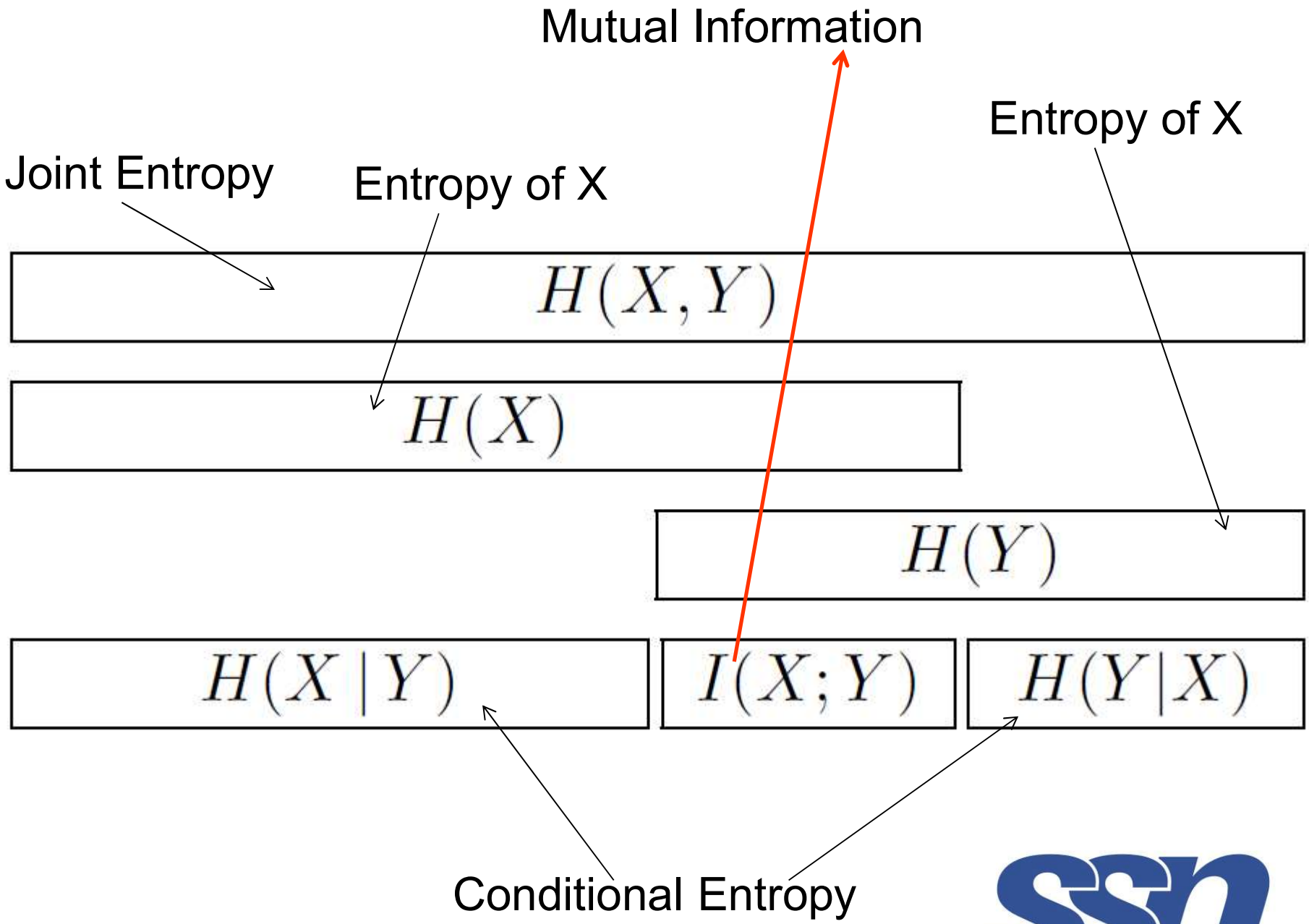
$$H(X,Y) = H(X) + H(Y \mid X) = H(Y) + H(X \mid Y)$$

Mutual Information

$$I(X;Y) \equiv H(X) - H(X \mid Y)$$

Entropy (information) about event X, after getting the knowledge of Y

Entropy (information) about event X, without the knowledge of Y


Mutual Information
Entropy of X
Joint Entropy
Entropy of X
$H(X,Y)$
$H(X)$
$H(Y)$
$H(X|Y)$
$I(X;Y)$
$H(Y|X)$
Conditional Entropy

Example – Find H(X,Y), H(X), H(Y), & H(X|Y)

$P(x,y)$		x			
		1	2	3	4
	1	$1/8$	$1/16$	$1/32$	$1/32$
y	2	$1/16$	$1/8$	$1/32$	$1/32$
	3	$1/16$	$1/16$	$1/16$	$1/16$
	4	$1/4$	0	0	0

Solution for H(X,Y)

Find out information content

	1	2	3	4
1	1/8	1/16	1/32	1/32
2	1/16	1/8	1/32	1/32
3	1/16	1/16	1/16	1/16
4	1/4	0	0	0

	1	2	3	4
1	3	4	5	5
2	4	3	5	5
3	4	4	4	4
4	2	0	0	0

Find out average information content

	1	2	3	4
1	1/8	1/16	1/32	1/32
2	1/16	1/8	1/32	1/32
3	1/16	1/16	1/16	1/16
4	1/4	0	0	0

X

	1	2	3	4
1	3	4	5	5
2	4	3	5	5
3	4	4	4	4
4	2	0	0	0

	1	2	3	4
1	3/8	4/16	5/32	5/32
2	4/16	3/8	5/32	5/32
3	4/16	4/16	4/16	4/16
4	2/4	0	0	0

Find out average information content

	1	2	3	4
1	3/8	4/16	5/32	5/32
2	4/16	3/8	5/32	5/32
3	4/16	4/16	4/16	4/16
4	2/4	0	0	0

Sum up

108/32 = 27/8 = 3.4 bits

Solution for H(X) and H(Y)

$P(x,y)$		x				$P(y)$	
		1	2	3	4		**I(y)**
y	1	$1/8$	$1/16$	$1/32$	$1/32$	$1/4$	2
	2	$1/16$	$1/8$	$1/32$	$1/32$	$1/4$	2
	3	$1/16$	$1/16$	$1/16$	$1/16$	$1/4$	2
	4	$1/4$	0	0	0	$1/4$	2
$P(x)$		$1/2$	$1/4$	$1/8$	$1/8$		

I(x)	**1**	**2**	**3**	**3**

Find average of H(X)

$P(x)$	$1/2$	$1/4$	$1/8$	$1/8$

X

I(x)	**1**	**2**	**3**	**3**
=	1/2	2/4	3/8	3/8

Sum up

14/8 = 7/4 = 1.75 bits

Find average of H(Y)

$P(y)$

$1/4$
$1/4$
$1/4$
$1/4$

X

I(y)	=
2	2/4
2	2/4
2	2/4
2	2/4

Sum up
2 bits

Solution for H(X|Y)

Recall, $P(x,y) = P(x|y).P(y)$

Solution for H(X|Y)

$P(x,y)$		x				$P(y)$
		1	2	3	4	
y	1	$1/8$	$1/16$	$1/32$	$1/32$	$1/4$
	2	$1/16$	$1/8$	$1/32$	$1/32$	$1/4$
	3	$1/16$	$1/16$	$1/16$	$1/16$	$1/4$
	4	$1/4$	0	0	0	$1/4$
$P(x)$		$1/2$	$1/4$	$1/8$	$1/8$	

$P(x \mid y)$		x			
		1	2	3	4
y	1	$1/2$	$1/4$	$1/8$	$1/8$
	2	$1/4$	$1/2$	$1/8$	$1/8$
	3	$1/4$	$1/4$	$1/4$	$1/4$
	4	1	0	0	0

Find out information content of (X|Y)

	1	2	3	4
1	1/2	1/4	1/8	1/8
2	1/4	1/2	1/8	1/8
3	1/4	1/4	1/4	1/4
4	1	0	0	0

	1	2	3	4
1	1	2	3	3
2	2	1	3	3
3	2	2	2	2
4	0	0	0	0

Find out average information content

	1	2	3	4
1	1/2	1/4	1/8	1/8
2	1/4	1/2	1/8	1/8
3	1/4	1/4	1/4	1/4
4	1	0	0	0

X

	1	2	3	4
1	1	2	3	3
2	2	1	3	3
3	2	2	2	2
4	0	0	0	0

	1	2	3	4
1	1/2	2/4	3/8	3/8
2	2/4	1/2	3/8	3/8
3	2/4	2/4	2/4	2/4
4	0	0	0	0

	1	2	3	4	(sum-up row wise) H(X|Y)	
1	1/2	2/4	3/8	3/8	14/8	← H(X|Y=1)
2	2/4	1/2	3/8	3/8	14/8	← H(X|Y=2)
3	2/4	2/4	2/4	2/4	2	← H(X|Y=3)
4	0	0	0	0	0	← H(X|Y=4)

- H(X) = 1.75 bits
- H(X|Y=4) = 0 & is less than H(X) - **learning decreases entropy**
- H(X|Y=3) = 2 & is greater than H(X) - **learning increases entropy**
- H(X|Y=2) = 1.75 & is equal to H(X) - **learning doesn't change entropy**
- H(X|Y=1) = 1.75 & is equal to H(X) - **learning doesn't change entropy**

	1	2	3	4	(sum-up row wise) H(X|Y)
1	1/2	2/4	3/8	3/8	14/8
2	2/4	1/2	3/8	3/8	14/8
3	2/4	2/4	2/4	2/4	2
4	0	0	0	0	0

Find average to get H(X|Y):

=P(Y=1). H(X|Y=1)+P(Y=2). H(X|Y=2)

+P(Y=3). H(X|Y=3)+P(Y=4). H(X|Y=4)

=(14/8).(1/4) + (14/8).(1/4) + 2.(1/4)+0

=14/32+14/32+2/4 = 44/32 = 11/8 = 1.37 bits

On an average, H(Y|X) < H(X) i.e. 1.37 < 1.75

Mutual Information

$$I(X;Y) \equiv H(X) - H(X|Y)$$

= 1.75 – 1.37 = 0.38 bits

- Compute H(Y|X) = 1.62 bits
- H(Y) = 2 bits
- I(X;Y) = H(Y) – H(Y|X) = 2- 1.62 = 0.38 bits